

Mathematical Analysis 2

Seminar 10

Line integrals (1)

$$(C): \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}, t \in [a, b] ; f: (C) \rightarrow \mathbb{R} \text{ the density}$$

$$\rightarrow \underbrace{\int_C f(x, y, z) ds}_{\text{the mass}} = \int_a^b f(x(t), y(t), z(t)) \cdot \underbrace{\sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt}_{ds}$$

Ex. 1 Compute $\int_C (x^2 + y^2) ds$,

$$(C) : \vec{r}(t) = \underbrace{a(t - \sin t)}_{x(t)} \cdot \vec{i} + \underbrace{a(1 - \cos t)}_{y(t)} \cdot \vec{j}, \quad t \in [0, 2\pi], \quad a > 0 \text{ fixed}$$

$$x'(t)^2 + y'(t)^2 = a^2 \left((1 - \cos t)^2 + \sin^2 t \right) = a^2 (1 - 2 \cos t + \underbrace{\cos^2 t + \sin^2 t}_{=1}) = a^2 (2 - 2 \cos t)$$

$$\frac{1 - \cos t}{2} = \sin^2 \frac{t}{2} \Rightarrow x'(t)^2 + y'(t)^2 = a^2 \cdot 4 \cdot \sin^2 \frac{t}{2} \Rightarrow ds = 2a \cdot \left| \sin \frac{t}{2} \right| dt = 2a \cdot \sin \frac{t}{2} dt$$

$\frac{t}{2} \in [0, \pi]$ $\xrightarrow{\quad}$

$$I = \int_0^{2\pi} a^2 \left((t - \sin t)^2 + (1 - \cos t)^2 \right) \cdot 2a \cdot \sin \frac{t}{2} dt = 2a^3 \int_0^{2\pi} \left(t^2 + \sin^2 t - 2t \sin t + \underbrace{1 + \cos^2 t}_{=2 - 2 \cos t} - 2 \cos t \right) \cdot \sin \frac{t}{2} dt$$

$$= 2a^3 \int_0^{2\pi} (t^2 + 2 - 2t \sin t - 2 \cos t) \sin \frac{t}{2} dt = \dots = 16a^3 \left(\pi^2 - \frac{16}{9} \right).$$

$\hookleftarrow$ individual study

Ex. 2 Compute $\int_C \mathbf{z} \, ds$,

$$(C) : \vec{r}(t) = \underbrace{t \cdot \cos t \cdot \vec{i}}_{x(t)} + \underbrace{t \cdot \sin t \cdot \vec{j}}_{y(t)} + \underbrace{t \cdot \vec{k}}_{z(t)},$$

$t \in [0, a]$, $a > 0$ fixed

$$x'(t)^2 + y'(t)^2 + z'(t)^2 = (\cos t - t \cdot \sin t)^2 + (\sin t + t \cdot \cos t)^2 + 1 = \underbrace{\cos^2 t}_{1} - \underbrace{2t \sin t \cdot \cos t}_{+ \quad 0} + \underbrace{t^2 \cdot \sin^2 t}_{+ \quad t^2} + \underbrace{\sin^2 t}_{1} + \underbrace{2t \sin t \cdot \cos t}_{+ \quad 0} + \underbrace{t^2 \cos^2 t}_{+ \quad t^2} + 1$$

$$ds = \sqrt{t^2 + 2} \, dt$$

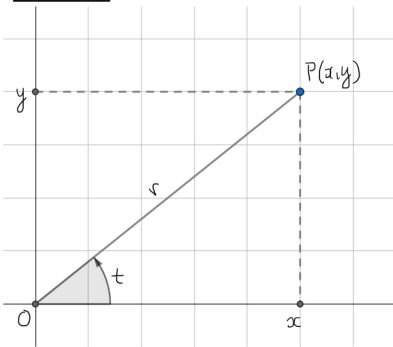
$$I = \int_0^a \underbrace{t}_{\vec{z}} \cdot \underbrace{\sqrt{t^2 + 2} \, dt}_{ds}$$

$$\begin{aligned} t^2 + 2 &= u \\ du &= 2t \, dt \end{aligned}$$

$$\frac{1}{2} \int_2^{a^2+2} u^{1/2} \, du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} \Big|_2^{a^2+2}$$

$$= \frac{1}{3} \left((a^2+2)^{3/2} - 2^{3/2} \right)$$

Ex. 3 Polar coordinates / polar parametrization of a plane curve:



(x, y) the Cartesian coordinates $\left| \begin{array}{l} x, y \in \mathbb{R} \\ r \geq 0, t \in [0, 2\pi] \text{ (or } t \in [-\pi, \pi]) \end{array} \right.$
 (r, t) the polar coordinates

$$\begin{cases} x = r \cdot \cos t \\ y = r \cdot \sin t \end{cases} \quad \begin{cases} r^2 = x^2 + y^2 \\ \tan(t) = \frac{y}{x} \end{cases}$$

(C) is a curve in the plane $\mapsto$ polar equation: $r = r(t)$

$$\mapsto \begin{cases} x(t) = r(t) \cos t \\ y(t) = r(t) \sin t \end{cases}$$

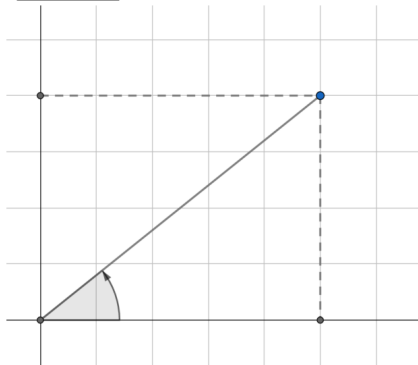
① the circle of radius R centered at $(0, 0)$
 $x^2 + y^2 = R^2 \iff \underline{r = R}$ (the polar equation) (constant)

$$\begin{cases} x(t) = R \cos t \\ y(t) = R \sin t \end{cases}$$

② The lemniscate of Bernoulli: $(x^2 + y^2)^2 = xy \iff r^4 = r^2 \cos t \cdot \sin t \iff r = 0$ or $r(t) = \sqrt{\cos t \cdot \sin t}$
 $\Rightarrow t \in [0, \frac{\pi}{2}]$ or $t \in [\pi, \frac{3\pi}{2}]$
 $x, y \geq 0$ $x, y \leq 0$ (the polar eq.) ≥ 0

$$\Rightarrow \begin{cases} x(t) = \sin^{\frac{1}{2}} t \cdot \cos^{\frac{3}{2}} t \\ y(t) = \sin^{\frac{3}{2}} t \cdot \cos^{\frac{1}{2}} t \end{cases}$$

Ex. 3 Polar coordinates / polar parametrization of a plane curve:



③ the cardioid:

$$(x^2 + y^2 - 2ax)^2 = 4a^2 \cdot (x^2 + y^2) \quad , a > 0$$

↓

$$(r^2 - 2a r \cdot \cos t)^2 = 4a^2 r^2$$

$$r^2 (r - 2a \cdot \cos t)^2 = 4a^2 r^2$$

$$r = 0 \quad \text{or} \quad |r - 2a \cdot \cos t| = 2a$$

$$r = 0 \quad \text{or} \quad r - 2a \cdot \cos t = \pm 2a$$

$$r = 0 \quad \text{or} \quad r = 2a \underbrace{(\cos t \pm 1)}_+$$

$$\Rightarrow r(t) = 2a(1 + \cos t) \quad (r=0 \text{ is obtained when } \cos t = -1)$$

e.g. $t = -\pi, t = \pi$

$$t \in [0, 2\pi] \quad \text{or} \quad t \in [-\pi, \pi]$$

$$\begin{cases} x(t) = 2a(1 + \cos t) \cdot \cos t \\ y(t) = 2a(1 + \cos t) \cdot \sin t \end{cases}$$